

Rova Isaia ANDRIAMAMY

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VA TER 14 Tsiadana Antananarivo, Madagascar

PROFESSIONAL OBJECTIVE

Master's student in Artificial Intelligence and Astrophysics with experience in Artificial Intelligence, data analysis, and data science. Passionate about machine learning and deep learning, I enjoy building practical, data-driven solutions for real-world challenges.

RESEARCH & PROJECT EXPERIENCE

• Deploying My First Machine Learning Model with Gradio

April 2026

Machine Learning, Data Analysis

Remotely

- Performed data analysis using **NumPy, Pandas, Matplotlib, and Seaborn**
- Applied data preprocessing (cleaning, normalization, categorical encoding, and feature engineering)
- Built and evaluated regression models using **MSE, RMSE, and R^2** metrics
- Deployed the machine learning model with **Gradio** and published on **Hugging Face Spaces**

• Development in Africa with Radio Astronomy (DARA) Training

July 2025 – March 2026

Radio & Optical Astronomy | Data Reduction & Analysis

South Africa/Kenya

- Mastered Bash scripting and Python libraries (NumPy, Astropy, SciPy, Matplotlib)
- Performed radio interferometry data reduction using CASA
- Hands on experience with radio/optical astronomy
- Collaboration and discussions with international professionals

• Hack4dev International Hackathon

April 2025

Deep Learning (Convolutional Neural Network)

Anxian University, Antananarivo

- Developed CNN architecture for classifying CubeSat imagery into **5 categories**
- Achieved **99.9% classification accuracy, 24× faster and 23× more compact** than the baseline model
- Ranked in the **Top 20 internationally**

• Food Price Forecasting in Madagascar

2024 – 2025

Deep Learning, Data Analysis

M1 Project

- Scraped and cleaned **18+ years** of market price data
- Engineered temporal features and Tested for RNN-based/LSTM/GRU models
- Achieved **94.51% accuracy** for LSTM
- Developed interactive Flask/React dashboard for real-time predictions

• BRICS (Brazil , Russia , India , China , South Africa) Astronomy Project

May 2025 – July 2025

Machine Learning, Data Analysis

Remotely

- Mastered Python libraries (pandas, NumPy, Matplotlib, Seaborn)
- Developed random forest and different models to predict halo masses
- Achieved **96.8% accuracy** with Random Forest Regressor model
- Scored **89/100** on final project evaluation

EDUCATION

• MASTER II <i>Astronomy and Astrophysics</i>	2025 – Present University of Antananarivo
• MASTER II <i>Artificial Intelligence</i>	2026 – Present INSI University of Ambanidia
• MASTER I <i>Artificial Intelligence</i>	2024 – 2025 INSI University of Ambanidia
• BACHELOR's Degree <i>Physics</i>	2022 – 2023 University of Antananarivo

TECHNICAL SKILLS

- **Programming Languages :** Python, C, R, MATLAB, HTML, CSS, JavaScript
- **Machine Learning & Deep Learning :** scikit-learn, PyTorch, TensorFlow (CNN, LSTM, RNN, objects Detection)
- **Data Analysis & Visualization :** Pandas, NumPy, Matplotlib, Seaborn
- **Frameworks & Cloud Technologies :** Flask, React, AWS
- **Database & Tools :** SQL, Power BI, Excel, Unix/Linux, Git, Docker

SELECTED CERTIFICATIONS

• English for Career Development  (English Training)	May 2026
• Deploying my first machine learning model with Gradio  (ML, Data Analysis)	April 2026
• Data Analysis with Power BI and Artificial Intelligence  (Power BI, Manus AI)	March 2026
• DARA Basic Training  (Data Reduction, Data Analysis)	March 2026
• Introduction to Data Visualization  (Power BI, DataWrapper, Google Sheet)	February 2026
• Introduction to Coding Through Play  (HTML, CSS, Python, Data Science)	February 2026
• Infrastructure for AI and Software Development  (Git, GitHub, Docker)	January 2026
• British English Pronunciation Practice  (Speaking Skills)	January 2026
• DARA Computer Training  (Linux Environment, Python)	August 2025
• BRICS Astronomy Project  (Data Analysis & Machine Learning)	July 2025
• American English Speaking Practice  (Communication Skills)	May 2025
• Hack4dev Hackathon  (Deep Learning, CNN)	April 2025

PROFESSIONAL MEMBERSHIPS

- **Malagasy Astronomical Society (MASS)** – Member participating
- **Benty Fields Astronomy Research** – Collaborative research group member
- **Sakay Young Students Association (AJES)** – Education volunteer and Activities
- **ASCUT** – University Sports and Cultural Association

ADDITIONAL INFORMATION

Languages: Malagasy (Native), French (Intermediate), English (Intermediate)
Interests: Data science, Data Analysis, Radio Astronomy

REFERENCES

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